

Fatty Liver — MASLD diagnosis & treatment apothecary



1. 7-10% weight loss

WHAT YOU SEE

Ladder sign: '>7-10% WT loss → stash ↓ fibrosis'

WHAT IT ENCODES

Sustained weight loss of 7-10% of body weight is the cornerstone of MASLD therapy and the single highest-yield intervention. At this magnitude it not only clears steatosis and resolves steatohepatitis but can REGRESS fibrosis — the outcome that matters for survival. The dose-response is steep: more weight loss yields more histologic benefit. Achieved via diet, exercise, and increasingly GLP-1 agonists or bariatric surgery.

BOARD TRAP

WRONG: expecting fibrosis regression from minimal (e.g., 3%) weight loss. Discriminator: ≥7-10% is required to improve fibrosis and resolve MASH; lesser losses help only steatosis — so the threshold for the meaningful (fibrosis) endpoint is 7-10%, not just 'some' weight loss.

2. 3-5% loss = steatosis

WHAT YOU SEE

Lower ladder sign: '3-5% WT loss → steatosis ↓'

WHAT IT ENCODES

A smaller weight loss of 3-5% improves hepatic STEATOSIS (fat content) but is generally insufficient to resolve steatohepatitis or reverse fibrosis. This stepwise dose-response is testable: 3-5% targets fat, ≥7% targets inflammation/MASH resolution, and ≥10% maximizes fibrosis regression. It frames realistic, graded goals for patients.

BOARD TRAP

WRONG: telling a MASH-with-fibrosis patient that 3-5% weight loss will reverse their fibrosis. Discriminator: 3-5% only reduces steatosis; resolving MASH needs ~7% and improving fibrosis needs ~10% — match the weight-loss target to the histologic goal.

3. Exercise

WHAT YOU SEE

Exercise bike glowing with '↓ fibrosis → exercise independent'

WHAT IT ENCODES

Exercise reduces hepatic fat and improves insulin sensitivity INDEPENDENT of weight loss — even without scale change, regular aerobic/resistance activity lowers liver fat. Guidelines suggest ~150-300 min/week of moderate activity. It complements dietary weight loss and is a low-risk, universally recommended pillar of MASLD care.

BOARD TRAP

WRONG: dismissing exercise as useless if a patient isn't losing weight. Discriminator: exercise lowers liver fat and improves metabolic health independent of weight change — recommend it regardless of whether the patient is achieving weight loss.

4. Mediterranean diet

WHAT YOU SEE

Plate of salad: 'best diet evidence'

WHAT IT ENCODES

The Mediterranean diet (high in olive oil, fish, nuts, vegetables, whole grains; low in refined carbohydrates, red meat, and especially fructose/sugar-sweetened beverages) has the strongest evidence base in MASLD. It reduces liver fat and improves insulin resistance and cardiovascular risk even independent of weight loss. Limiting fructose and sugary drinks is a specific, testable dietary point.

BOARD TRAP

WRONG: recommending a generic low-fat diet over the Mediterranean pattern. Discriminator: the Mediterranean diet (not low-fat) has the best MASLD evidence; cutting fructose/sugar-sweetened beverages and refined carbs is more important than simply lowering total fat.

5. Coffee 3 cups/day

WHAT YOU SEE

Coffee cups: '3 cups/day ↓ fibrosis ↓ HCC'

WHAT IT ENCODES

Coffee consumption (≥3 cups/day, caffeinated) is associated with reduced fibrosis progression and lower hepatocellular carcinoma risk across chronic liver diseases, including MASLD. The hepatoprotective effect is observational but consistent and is a favorable lifestyle point. It is a recognized 'safe to encourage' adjunct.

BOARD TRAP

WRONG: advising MASLD patients to avoid coffee to 'protect the liver.' Discriminator: coffee is hepatoprotective — ≥3 cups/day is associated with less fibrosis and lower HCC risk, so it should be encouraged, not restricted (the harmful beverages are sugar-sweetened/fructose drinks).

6. Pioglitazone

WHAT YOU SEE

Jar 'PIOGLITAZONE PPAR-γ', plus 'MASH + DM2' tag

WHAT IT ENCODES

Pioglitazone, a PPAR-γ agonist (thiazolidinedione) that improves insulin sensitivity, improves NASH/MASH histology including some fibrosis benefit and is favored when the patient has concurrent type 2 diabetes. Key adverse effects to counsel: weight gain, fluid retention/heart failure exacerbation, increased fracture risk, and possible bladder-cancer signal. It is an established off-label option for biopsy-proven MASH.

BOARD TRAP

WRONG: prescribing pioglitazone to a MASH patient with heart failure or osteoporosis. Discriminator: pioglitazone causes fluid retention (avoid in NYHA III-IV heart failure) and fracture risk — it improves MASH histology but is contraindicated/cautioned in those comorbidities; best suited to MASH + type 2 diabetes without HF.

7. Resmetirom

WHAT YOU SEE

Bottle 'RESMETIROM THR-β FDA 2024'

WHAT IT ENCODES

Resmetirom is a liver-directed thyroid hormone receptor-beta (THR-β) agonist that increases hepatic fat metabolism, and in March 2024 it became the FIRST FDA-approved drug specifically for MASH. It is indicated for noninvasively or biopsy-confirmed MASH with moderate-to-advanced fibrosis (F2-F3), alongside diet and exercise, and showed both MASH resolution and fibrosis improvement in the MAESTRO trial. It is the marquee new-drug answer for this topic.

BOARD TRAP

WRONG: naming an older agent as the 'first FDA-approved MASH drug.' Discriminator: resmetirom (THR-β agonist, 2024) is the first FDA-approved therapy for MASH; vitamin E, pioglitazone, and GLP-1 agonists are used off-label/for comorbidities — only resmetirom carries the specific MASH-F2/F3 approval.

8. MASH F2-F3 target

WHAT YOU SEE

A tag on the resmetirom bottle reading 'MASH F2-F3' — it marks the moderate-to-advanced-fibrosis population that the drug actually targets.

WHAT IT ENCODES

Pharmacologic therapy with resmetirom targets confirmed MASH with MODERATE-to-ADVANCED fibrosis (stage F2-F3) — the population at meaningful risk of progression where drug benefit outweighs risk. Patients with simple steatosis or minimal fibrosis (F0-F1) are managed with lifestyle alone, and decompensated cirrhosis (F4 with decompensation) was excluded from approval. Match the drug to the F2-F3 'at-risk MASH' niche.

BOARD TRAP

WRONG: starting resmetirom in bland steatosis or in decompensated cirrhosis. Discriminator: the approved/indicated population is non-cirrhotic MASH with F2-F3 fibrosis; F0-F1 gets lifestyle only, and decompensated F4 was excluded — drug selection is fibrosis-stage-driven.

9. FIB-4 formula

WHAT YOU SEE

Banner 'Age × AST ÷ Platelet × √ALT'

WHAT IT ENCODES

FIB-4 = (Age × AST) / (Platelets × √ALT) is the recommended FIRST-LINE noninvasive test to risk-stratify fibrosis in MASLD, using only routine labs. It exploits the facts that fibrosis raises AST and lowers platelets (from portal hypertension/splenic sequestration). A normal FIB-4 has a high negative predictive value, letting clinicians avoid biopsy in most patients. It is the gateway to the staged diagnostic pathway.

BOARD TRAP

WRONG: jumping to liver biopsy as the first step to assess fibrosis. Discriminator: the first-line tool is the noninvasive FIB-4 (then elastography for intermediate/high scores); biopsy is reserved for diagnostic uncertainty — start with FIB-4, not biopsy.

10. FIB-4 <1.3 low risk

WHAT YOU SEE

Green bottle '<1.3' with checkmark, 'no biopsy needed'

WHAT IT ENCODES

A FIB-4 <1.3 has a high negative predictive value for advanced fibrosis, effectively ruling it out — these low-risk patients need NO biopsy and no specialist referral, just metabolic management and periodic reassessment (e.g., every 1-3 years). This is the high-volume primary-care triage cutoff. (Use a higher threshold ~2.0 in patients over 65 to avoid false positives.)

BOARD TRAP

WRONG: ordering elastography or biopsy in a MASLD patient with FIB-4 <1.3. Discriminator: <1.3 rules out advanced fibrosis (high NPV) — manage risk factors and recheck periodically; no biopsy or hepatology referral is needed for these low-risk patients.

11. FIB-4 >2.67 advanced

WHAT YOU SEE

Red bottle '>2.67 → advanced fibrosis'

WHAT IT ENCODES

A FIB-4 >2.67 suggests a high probability of advanced fibrosis and warrants referral to hepatology and confirmatory vibration-controlled transient elastography (FibroScan) or MR elastography. Scores between 1.3 and 2.67 are indeterminate and also proceed to elastography. This tiered approach (FIB-4 → elastography → selective biopsy) minimizes unnecessary biopsies.

BOARD TRAP

WRONG: reassuring or discharging a patient with FIB-4 >2.67. Discriminator: >2.67 flags likely advanced fibrosis needing elastography and hepatology referral; the indeterminate 1.3-2.67 zone is NOT low-risk either and also requires further (elastography) testing.

12. Metformin no benefit

WHAT YOU SEE

Metformin bottle with red X 'no histological benefit'

WHAT IT ENCODES

Metformin does NOT improve MASH/NASH histology or fibrosis and is not a MASLD-specific treatment, despite being first-line for the diabetes these patients often have. Use it to manage type 2 diabetes if indicated, but do not credit it with treating the liver disease. It is a classic distractor because patients are frequently already on it.

BOARD TRAP

WRONG answer: selecting metformin to treat MASH/fibrosis. Discriminator: metformin has no proven histologic liver benefit — the histology-modifying options are pioglitazone, vitamin E (non-diabetic), GLP-1 agonists, and resmetirom; metformin only addresses concurrent diabetes.

13. Statins safe

WHAT YOU SEE

Statin bottle, X over 'MASH treatment' but green 'safe in MASLD + CVD benefit'

WHAT IT ENCODES

Statins do NOT directly treat MASH but are SAFE in MASLD — including with mildly elevated transaminases and in compensated cirrhosis — and are strongly indicated because cardiovascular disease is the LEADING cause of death in MASLD patients. They reduce CV events and may lower portal pressure and HCC risk. Withholding a statin over mild ALT elevation is a frequent and tested error.

BOARD TRAP

WRONG: stopping a statin in a MASLD patient with modestly elevated liver enzymes. Discriminator: statins are safe and indicated in MASLD (CVD is the #1 killer); mild transaminase elevation is not a contraindication — continue the statin rather than stopping it 'to protect the liver.'

14. Vit E / not proven

WHAT YOU SEE

A cluster of supplement bottles (including vitamin E) stamped with red Xs and 'not proven' — only vitamin E in non-diabetic MASH has limited evidence; the rest are unproven distractors.

WHAT IT ENCODES

Vitamin E (alpha-tocopherol, ~800 IU/day) can improve steatohepatitis histology in NON-diabetic adults with biopsy-proven MASH (PIVENS trial) and is an option in that specific group; however, it lacks proven fibrosis benefit and is not recommended in diabetics or as universal therapy. High-dose vitamin E carries possible risks (hemorrhagic stroke, prostate cancer signal). Most other 'supplements' depicted here are unproven and not standard.

BOARD TRAP

WRONG: prescribing vitamin E broadly to diabetic MASH patients or expecting it to reverse fibrosis. Discriminator: vitamin E's evidence is limited to NON-diabetic biopsy-proven MASH and to inflammation/steatosis (not fibrosis); it's not first-line for everyone and is avoided in diabetics — other supplements remain unproven.

15. Bariatric surgery

WHAT YOU SEE

Scale 'BARIATRIC SURGERY 60-80% fibrosis improvement → 1 year'

WHAT IT ENCODES

Bariatric (metabolic) surgery produces the most dramatic and durable results in eligible obese MASLD patients — roughly 60-80% achieve fibrosis improvement and high rates of MASH resolution within about a year, by driving large sustained weight loss and metabolic correction. It is an option for those meeting BMI criteria, and is generally avoided in decompensated cirrhosis. It outperforms current pharmacotherapy on histologic endpoints.

BOARD TRAP

WRONG: offering bariatric surgery to a patient with decompensated cirrhosis/portal hypertension. Discriminator: bariatric surgery markedly improves MASH/fibrosis in obese patients but is high-risk and generally contraindicated in decompensated cirrhosis — reserve it for those meeting BMI criteria without advanced decompensation.

16. BMI thresholds

WHAT YOU SEE

Sign 'BMI >40 OR >35 + comorbidities'

WHAT IT ENCODES

Traditional bariatric-surgery criteria are BMI ≥ 40 , OR BMI ≥ 35 with an obesity-related comorbidity (type 2 diabetes, hypertension, obstructive sleep apnea, MASLD) — and newer guidelines extend consideration to BMI ≥ 30 with poorly controlled diabetes. MASLD/MASH itself counts as a qualifying comorbidity at the ≥ 35 threshold. Know the ≥ 40 / ≥ 35 -plus-comorbidity numbers.

BOARD TRAP

WRONG: requiring BMI ≥ 40 for every bariatric candidate. Discriminator: BMI ≥ 35 WITH a comorbidity (and MASH counts) also qualifies — and contemporary criteria reach down to BMI ≥ 30 with uncontrolled diabetes; the ≥ 40 cutoff is not the only entry point.